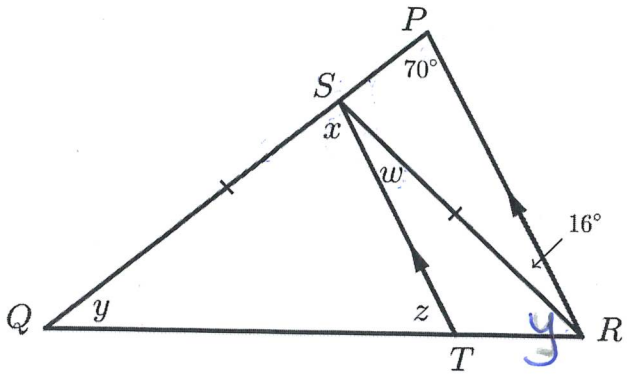
Memo - Geometry Revision

10.2 In the diagram $ST \parallel PR$ and $SR = SQ$. $\hat{P} = 70°$ and $P\hat{R}S = 16°$.



Determine, with reasons, the size of angles w, x, y, and z. (7)

$w = P\hat{R}S = 16°$ alt ∠s : ST // PR

$x = \hat{P} = 70°$ corresp ∠s : ST // PR

In ΔSQR, $\hat{Q} = \hat{R} = y$ (isos Δ : SR = SQ)

$\therefore y + x + w + y = 180°$ ∠s of Δ

$2y + 70° + 16° = 180°$

$2y = 94$

$y = 47°$

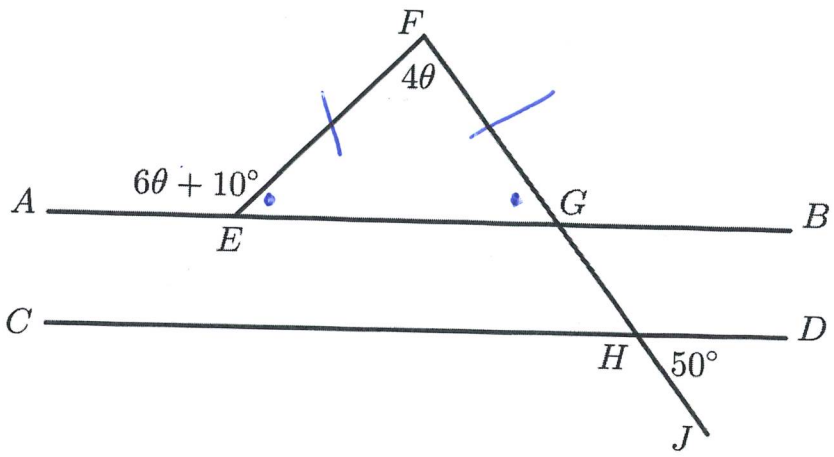
$z = 180° - x - y$ (∠s of ΔQST)

$= 180° - 70° - 47°$

$z = 63°$

10.3 In the given figure, $AEGB$, $FGHJ$, and CHD are straight lines. $FE = FG$

$A\hat{E}F = 6\theta + 10^\circ$; $\hat{F} = 4\theta$; $J\hat{H}D = 50^\circ$



Prove with full reasoning that $AB \parallel CD$. (7)

$F\hat{G}E + \hat{F} = 6\theta + 10^\circ$ ext $\angle$ of $\triangle FEG$

$F\hat{G}E + 4\theta = 6\theta + 10^\circ$

$F\hat{G}E = 2\theta + 10^\circ$ } $\angle$s opp = sides

$\therefore F\hat{E}G = 2\theta + 10^\circ$ }

$2\theta + 10^\circ + 2\theta + 10^\circ + 4\theta = 180^\circ$ ($\angle$s of $\triangle$)

$8\theta + 20^\circ = 180^\circ$

$8\theta = 160$

$\theta = 20^\circ$

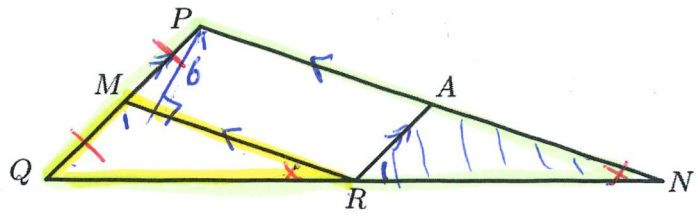
$\therefore F\hat{G}E = 2(20) + 10^\circ = 50^\circ$

$J\hat{H}D = G\hat{H}C = 50^\circ$ Vert opp $\angle$s

$\therefore G\hat{H}C = F\hat{G}E = 50^\circ$

$AB \parallel CD$ (converse corresponding $\angle$s)

10.4 In the figure, $PARM$ is a parallelogram and $QM = PM$.



10.4.1 Prove, with reasons, that $\triangle QMR \equiv \triangle QPN$. (4)

10.4.2 Prove, with reasons, that $\triangle NRA \equiv \triangle RQM$. (4)

10.4.3 If the perpendicular distance from P to MR is $6cm$ and the area of the parallelogram is $72cm^2$, determine the length of AN. (4)

10.4.1 In $\triangle QMR$ & $\triangle QPN$

1. $\hat{Q}$ is common
2. $\hat{M}_1 = \hat{P}$ corresp ∠s: MR // PN
3. $\hat{R} = \hat{N}$ ∠s of △s

$\triangle QMR /// \triangle QPN$ (AAA)

10.4.2 In $\triangle NRA$ & $\triangle RQM$

1. $\hat{Q} = \hat{R}_1$ corresp ∠s: PQ // RA
2. $\hat{R} = \hat{N}$ Proved in 10.4.1
3. PM = RA opp sides parm

But PM = QM given

∴ QM = RA

∴ $\triangle NRA \equiv \triangle RQM$ (SAA)

10.4.3 AN = RM sides of ≡ △'s

Area of Parm = base × ht

= RM × 6

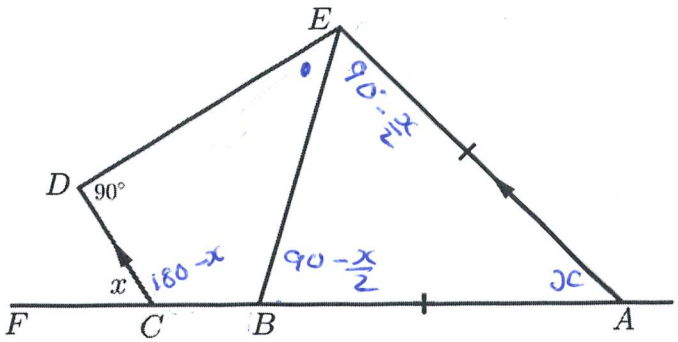
72 = 6RM

12 = RM

∴ AN = 12cm

QUESTION 11

11.1 In the figure $AE \parallel CD$, $C\hat{D}E = 90°$, $AE = AB$, and $D\hat{C}F = x$



Determine with full reasons, the value of $B\hat{E}D$ in terms of x. (5)

$\hat{A} = x$ corresp ∠'s: DC // AE

$B\hat{E}A = E\hat{B}A = \frac{180° - x}{2}$ ∠SOS Δ : AE = AB

$= 90° - \frac{x}{2}$

$D\hat{C}B = 180° - x$ ∠'s on str line

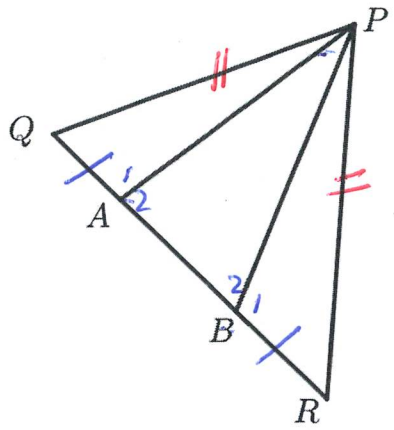
$\hat{D} + D\hat{E}B + B\hat{E}A + \hat{A} + D\hat{C}B = 360$ ∠'s of quad

$90° + D\hat{E}B + 90 - \frac{x}{2} + x + 180° - x = 360°$

$360° + D\hat{E}B - \frac{x}{2} = 360°$

$D\hat{E}B = \frac{x}{2}$

11.4 In the figure, A and B lie on side QR of ΔPQR such that $QA = RB$. $PQ = PR$.



11.4.1 Prove, with reasons, that $\Delta APQ \equiv \Delta BPR$. (4)

11.4.2 Prove, with reasons, that $P\hat{A}B = P\hat{B}A$. (3)

11.4.1 In ΔAPQ & ΔBPR

1. $\hat{Q} = \hat{R}$ isos Δ $PQ = PR$
2. $PQ = PR$ Given
3. $QA = RB$ Given

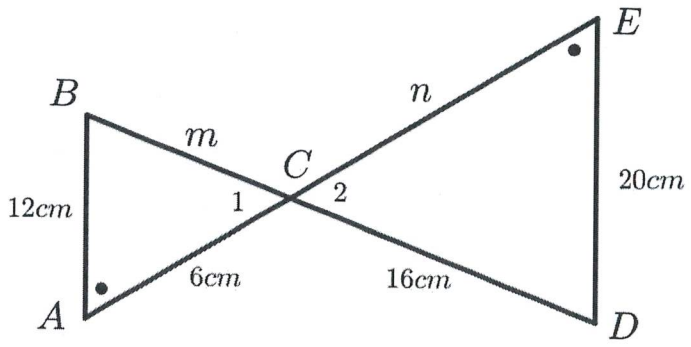
$\Delta APQ \equiv \Delta BPR$ (SAS)

11.4.2 $P\hat{A}Q = P\hat{R}B$ $\angle$s of $\equiv \Delta$s

$\hat{A}_1 + \hat{A}_2 = \hat{B}_1 + \hat{B}_2$ $\angle$s on str line

$\therefore \hat{A}_2 = \hat{B}_2$

11.5 In the figure, $\hat{A} = \hat{E}$.



11.5.1 Prove that $\Delta ABC \;|||\; \Delta EDC$. (4)

11.5.2 Hence determine the values of m and n. (4)

11.5.1 In $\triangle ABC$ & $\triangle EDC$

1. $\hat{C}_1 = \hat{C}_2$ Vert opp $\angle$'s
2. $\hat{A} = \hat{E}$ Given
3. $\hat{B} = \hat{D}$ Sum $\angle$'s of $\triangle$.

$\therefore \triangle ABC \;///\; \triangle EDC$ (AAA)

11.5.2 $\frac{AB}{ED} = \frac{BC}{DC} = \frac{AC}{EC}$ sides in prop

$\frac{12}{20} = \frac{m}{16} = \frac{6}{n}$

$\frac{12}{20} = \frac{m}{16}$

$20m = 192$

$m = 9{,}6\text{cm}$

$\frac{12}{20} = \frac{6}{n}$

$12n = 120$

$n = 10$